

## Management of Urinary Tract Infections

| Questions                                                                                                                                                                                                                                                             | Please answer here                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | For pharmacist                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gender:</b> What is the sex of the patient?                                                                                                                                                                                                                        | Female / Male or individuals who have undergone gender reassignment surgery                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | [Triggers: Male or individuals who have undergone gender reassignment surgery → <b>Provide usual care and/or refer to GP</b> ]                                                                                                                                                                                                                                   |
| <b>Age:</b> Is the patient aged between 18 and 65 years?                                                                                                                                                                                                              | Age ..... years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | [Triggers: No (under 18 or over 65) → <b>Provide usual care and/or refer to GP</b> ]                                                                                                                                                                                                                                                                             |
| <b>Consent:</b> Does the patient consent? *<br>(*The patient must be physically present in the pharmacy)                                                                                                                                                              | No / Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | [Triggers: No → <b>Provide usual care and/or refer to GP</b> ]                                                                                                                                                                                                                                                                                                   |
| <b>Symptoms:</b> Is the patient presenting with 2 or more symptoms of acute cystitis?*<br>*If vaginal discharge is absent and at least two symptoms of cystitis are present (i.e. acute dysuria, frequency, urgency), the probability of cystitis is greater than 90% | No / Yes<br><ul style="list-style-type: none"> <li>○ Dysuria</li> <li>○ Urinary frequency</li> <li>○ Urinary urgency</li> <li>○ Suprapubic pain or discomfort</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | [Triggers: No (presenting with only 1 symptom of acute cystitis or atypical symptoms) → <b>Provide usual care and refer to GP</b> as patient may benefit from laboratory investigations]                                                                                                                                                                         |
| <b>Differential diagnosis:</b> Do symptoms of medical history suggest a cause other than acute cystitis                                                                                                                                                               | Yes / No<br>(e.g. <b>vaginal discharge, or vulval and vaginal discomfort</b> ) → (e.g. <b>vaginal thrush, bacterial vaginosis, vaginitis or vulvitis, chlamydia, gonorrhoea, trichomoniasis</b> or other)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | [Triggers: Consider <b>appropriateness of S3 Pharmacist Only thrush treatment</b> or <b>refer to GP</b> for investigation of other causes (e.g. infections such as bacterial vaginosis or herpes, or skin conditions)]                                                                                                                                           |
| <b>Red flag Symptoms:</b> Does the patient have signs and symptoms indicating <b>pyelonephritis</b> ? (e.g. <b>fever &gt;38°C, chills, nausea, vomiting, flank pain</b> )                                                                                             | No / Yes<br><ul style="list-style-type: none"> <li>○ <b>Fever &gt;38°C</b></li> <li>○ <b>Chills</b></li> <li>○ <b>Nausea</b></li> <li>○ <b>Vomiting</b></li> <li>○ <b>Flank pain</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | [Triggers: Yes → <b>Immediate referral to Emergency Department</b> ]                                                                                                                                                                                                                                                                                             |
| <b>Reg flag medical Hx:</b> Does the patient have any of the following?                                                                                                                                                                                               | No / Yes<br><ul style="list-style-type: none"> <li>○ <b>Pregnancy</b></li> <li>○ Less than <b>6 weeks post-partum</b></li> <li>○ Gross <b>haematuria</b></li> <li>○ <b>Cerebral palsy</b> or other neurological diseases that may affect bladder function</li> <li>○ Recent insertion of <b>IUD (within the last 3 months)</b></li> <li>○ Urinary <b>catheterisation (in situ or in the last 48 hours)</b></li> <li>○ Presence of history of <b>anatomical or functional abnormalities of the urinary tract</b> including: (Urinary tract obstruction, Urinary tract abnormality, Urolithiasis, Nephrostomy tube, Ureteral stent, Spinal cord injury)</li> </ul>                                                                                                      | [Triggers: Yes → <b>Immediate referral to GP</b> ]<br><b>Symptomatic treatment</b> with nonprescription medicines (if appropriate) and <b>self-care advice</b> may be provided by pharmacist prior to review by GP                                                                                                                                               |
| <b>Risks:</b> Does the patient report any of the following?                                                                                                                                                                                                           | No / Yes<br><ul style="list-style-type: none"> <li>○ Having <b>diabetes</b> or using <b>medicines that increase the risk of UTI</b> (e.g. SGLT2 inhibitors)</li> <li>○ <b>Reoccurrence of UTI symptoms within 2 weeks</b> of completing appropriate antimicrobial treatment for a previous episode</li> <li>○ Symptoms of UTI <b>persisting 48-72 hours</b> after starting appropriate antibiotic treatment</li> <li>○ <b>2 or more</b> symptomatic UTIs in previous 6 months OR <b>3 or more</b> in previous 12 months</li> <li>○ Long-term <b>inpatient care</b> (including residential aged care facilities)</li> </ul>                                                                                                                                            | [Triggers: Yes → <b>Provide usual care and refer to GP</b> as patient may benefit from laboratory investigations]                                                                                                                                                                                                                                                |
| <b>Soft triggers:</b> Does the patient report / present with any of the following?                                                                                                                                                                                    | No / Yes<br><ul style="list-style-type: none"> <li>○ Risk factors for <b>sexually transmitted infection (STI)</b> including: <ul style="list-style-type: none"> <li>▪ Age &lt; 30 years</li> <li>▪ Previous STI</li> <li>▪ Sexual contact without a condom/dental dam outside a mutually monogamous relationship</li> <li>▪ A new sexual partner in last 60 days</li> <li>▪ A sexual partner recently treated for an STI</li> <li>▪ Sexual contact with a sex worker</li> </ul> </li> <li>○ History of <b>pyelonephritis</b></li> <li>○ Taking <b>immunosuppressant medicines</b></li> <li>○ Being <b>immunocompromised</b></li> <li>○ <b>Asplenia</b></li> <li>○ <b>Renal disease</b> or impairment</li> <li>○ <b>IUD in situ</b> (greater than 3 months)</li> </ul> | [Triggers: Yes → <b>Provide usual care and refer to GP</b> ] (** Refer patient for STI testing where risk factors are identified.)<br>→ concurrent to a referral to the GP, if clinically appropriate (e.g. <b>possible delay in access to GP</b> and no contraindications to antibiotics). → <b>Antibiotic treatment by pharmacist may still be considered.</b> |

\*\*\* **Provide usual care** = including symptomatic treatment with non-prescription medicines (if appropriate) and self-care advice

## PHARMACIST CARE

|                         |                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONSERVATIVE MANAGEMENT | non-prescription medicines (as first line treatment for patients without immune compromise) | <ul style="list-style-type: none"> <li>Analgesia: <b>ibuprofen 400mg orally, every 8 hours for up to 3 days</b>. Maximum daily dose of 2.4g in 24 hours</li> </ul>                                                                                                                                                                                                                                                          | If patient continues to have symptoms <b>after 48 hours</b> of conservative management, advise patient to return to pharmacy for review and consider <b>antibiotic therapy</b> . |
|                         | non-pharmacological and self-care advice                                                    | <ul style="list-style-type: none"> <li>Increasing <b>water intake (up to 1.5 L daily)</b> may reduce the risk of recurrent UTI in premenopausal females who have inadequate fluid intake (&lt;1.5 L per day)</li> <li><b>CMI and/or a Self-Care Fact Card</b></li> <li>guidance on recognising signs and symptoms of <b>worsening infection (e.g. fever 38°C or higher, rigors, loin or back pain, vomiting)</b></li> </ul> |                                                                                                                                                                                  |

\*\*The efficacy of **urinary alkalinising** agents for the symptomatic treatment of acute cystitis has not been established in trials, but some patients find them useful for symptom relief.

\*\***Cranberry products, ascorbic acid and methenamine hippurate** are not effective for the treatment of acute cystitis.

## EMPIRICAL ANTIBIOTIC THERAPY

|                                                                                                                                                                                                                      |                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ANTIBIOTIC THERAPY</b><br><br><b>Empirical antibiotic therapy, where indicated</b><br>Dispense any medications (if supplied) via pharmacy dispensing software and label according to the legislative requirements | <b>Nitrofurantoin (1st line)</b> 100mg orally, every 6 hours for 5 days (Supply 20 capsules)  | <b>Contraindications:</b> <ul style="list-style-type: none"> <li>Previous serious adverse reaction to nitrofurantoin</li> <li><b>G6PD</b>, enolase, or glutathione peroxidase deficiency (may lead to haemolytic anaemia)</li> <li>Severe <b>renal impairment</b></li> <li>Avoid in <b>breastfeeding</b> if infant is under one month or has G6PD deficiency</li> </ul>                                                                                                                                                                                                                                                                                                                                      | <b>Provide follow-up advice</b> and expectations around duration of symptoms <ul style="list-style-type: none"> <li>Symptoms should respond to appropriate antibiotic treatment within 48 hours.</li> <li>If symptoms <b>persist 48-72 hours after finishing antibiotic</b> treatment or symptoms develop that are not symptoms of acute cystitis, patient should be advised to <b>see a GP</b>.</li> </ul> |
|                                                                                                                                                                                                                      | <b>Fosfomycin (2nd line)</b> 3g orally, as a single dose (at night) (Supply 1 sachet)         | <b>Contraindications:</b> <ul style="list-style-type: none"> <li>Previous serious adverse reaction or hypersensitivity to Fosfomycin</li> <li>Severe <b>renal impairment</b></li> <li>Not recommended in cases of fructose intolerance, glucose-galactose malabsorption or sucrase-isomaltase insufficiency</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                                                                                                                                                                      | <b>Trimethoprim (3rd line)</b> 300mg orally, daily (at night) for 3 nights (Supply 3 tablets) | <b>Contraindications:</b> <ul style="list-style-type: none"> <li>If the patient has been treated with trimethoprim <b>in the previous 3 months</b>, or had a trimethoprim-resistant E. coli isolate during this time, use an alternative antibiotic for empirical therapy.</li> <li>Previous serious adverse reaction to trimethoprim-containing medicines</li> <li>Megaloblastic <b>anaemia due to folate deficiency</b></li> <li>Other causes of <b>folate deficiency</b></li> <li>Other severe blood disorders</li> <li>Porphyria</li> <li><b>Hyperkalaemia</b></li> <li>Treatment with <b>methotrexate</b></li> <li>Treatment with <b>phenytoin</b></li> <li>Treatment with <b>lamivudine</b></li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                             |

Patients on certain antibiotics (e.g., **nitrofurantoin and fosfomycin**) should **avoid the use of alkalinising agents** as they may significantly reduce the efficacy of their antibiotics.